

Experimental Workflow for Solar Flare Prediction

HMI active-region observations

950,047 observations | 1,570 active regions | 29 engineered numeric features

Active-region grouped split

Train / Validation / Test: 759,357 / 95,933 / 94,757 observations

IMAGE-SEQUENCE BRANCH

10-frame chronological image sequences

CNN-LSTM | multiclass C / M / X

Current training: focal loss (gamma 0.8)
inverse-square-root class sampling + class weights
Checkpoint: cnn_lstm_best.pth
GPU inference: PyTorch 2.11.0 + ROCm 7.2

Class probabilities

2,021 test sequences | C / M / X

TABULAR BRANCH

29 engineered numeric features; separate table

Tabular classifiers

XGBoost | Logistic Regression
HistGradientBoosting | Extra Trees
ge_c: C/M/X vs 0 | ge_m: M/X vs 0/C

Event probabilities

Validation threshold frozen for Test

Held-out Test evaluation

Separate task-specific predictions; no matched comparison

ROC / PR analysis

From task-specific probabilities

Calibration

Tabular Brier score / ECE

Threshold metrics

Tabular POD / FAR / CSI / F1
TSS / HSS / MCC

2,000-replicate active-region cluster bootstrap | 95% CIs

Tabular thresholds: Validation maximum TSS over 0.1-0.9; frozen for Test.
Image and engineered-feature pipelines are complementary, non-matched evaluation tasks.